THE STACK OF RELATIVELY PERFECT COMPLEXES IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5493. Standing situation: X -> B and the relatively perfect K.
- Lemma lemma-complexes-diagonal (canonical tag 0DLG). cited at proof line 5502. The diagonal of the complexes stack is representable.
- Lemma algebraic-lemma-representable-diagonal (canonical tag 045G). cited at proof lines 5502-5503. Representable diagonal makes Isom a sheaf.
- Lemma spaces-descent-lemma-fpqc-universal-effective-epimorphisms (canonical tag 04P2). cited at proof lines 5517-5518. The morphisms g_i glue uniquely to g.
- Lemma lemma-complexes (canonical tag 0DLD). cited at proof lines 5539-5540 and 5559. Negative Ext vanishing enables descent; condition (2) characterized.
- Lemma spaces-simplicial-lemma-fppf-glue-neg-ext-zero (canonical tag 0DLA). cited at proof lines 5541 and 5548 (footnote). Descent of complexes along the Cech fppf hypercovering.
- Lemma spaces-more-morphisms-lemma-pseudo-coherent-descends-fpqc (canonical tag 0DL1). cited at proof lines 5555-5556. Pseudo-coherence of the descended complex.
- Lemma spaces-more-morphisms-lemma-tor-amplitude-descends-fppf (canonical tag 0DL2). cited at proof lines 5556-5557. Tor amplitude descends.
- Lemma lemma-complexes-open-neg-exts-vanishing (canonical tag 0DLC). cited at proof lines 5560-5561. Open W verifies condition (2) for the descended object.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5493: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 0DLG: sources/stacks-chow/tags.txt, source lines 17046–17046. Cited at proof lines 5502: Isom presheaves are sheaves via the diagonal; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 045G: sources/stacks-chow/tags.txt, source lines 5435–5435. Cited at proof lines 5502-5503: the transfer lemma from diagonal to Isom sheaves; load-bearing.

Source reference. Original lemma 04P2: sources/stacks-chow/tags.txt, source lines 6100–6100. Cited at proof lines 5517-5518: g_i glue uniquely to g : T -> B; load-bearing.

Source reference. Original lemma 0DLD: sources/stacks-chow/tags.txt, source lines 17043–17043. Cited at proof lines 5539-5540 (vanishing enabling descent) and 5559 (condition (2) for the descended E); load-bearing twice.

Source reference. Original lemma 0DLA: sources/stacks-chow/tags.txt, source lines 17040–17040. Cited at proof lines 5541 and 5548 (footnote Cech hypercovering): descent for the complexes; load-bearing twice.

Source reference. Original lemma 0DL1: sources/stacks-chow/tags.txt, source lines 17031–17031. Cited at proof lines 5555-5556: E is pseudo-coherent; load-bearing.

Source reference. Original lemma 0DL2: sources/stacks-chow/tags.txt, source lines 17032–17032. Cited at proof lines 5556-5557: locally finite tor dimension descends, so E is T-perfect; load-bearing.

Source reference. Original lemma 0DLC: sources/stacks-chow/tags.txt, source lines 17042–17042. Cited at proof lines 5560-5561: the description of the open W verifies condition (2) for E; load-bearing.

Lemma 0.2. *In Situation 0DLE the functor $p : \mathcal{C}omplexes_{X/B} \longrightarrow (Sch/S)_{fppf}$ is a stack in groupoids.*

Proof. To prove that $\mathcal{C}omplexes_{X/B}$ is a stack in groupoids, we have to show that the presheaves *Isom* are sheaves and that descent data are effective. The statement on *Isom* follows from Lemma 0DLG, see Algebraic Stacks, Lemma 045G. Let us prove the statement on descent data.

Suppose that $\{a_i : T_i \to T\}$ is an fppf covering of schemes over S. Let (ξ_i, φ_{ij}) be a descent datum for $\{T_i \to T\}$ with values in $\mathcal{C}omplexes_{X/B}$. For each i we can write $\xi_i = (T_i, g_i, E_i)$. Denote $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ and $\mathrm{pr}_1 : T_i \times_T T_j \to T_j$ the projections. The condition that $\xi_i|_{T_i \times_T T_j} \cong \xi_j|_{T_i \times_T T_j}$ implies in particular that $g_i \circ \mathrm{pr}_0 = g_j \circ \mathrm{pr}_1$. Thus there exists a unique morphism $g : T \to B$ such that $g_i = g \circ a_i$, see Descent on Spaces, Lemma 04P2. Denote $X_T = T \times_{g,B} X$. Set $X_i = X_{T_i} = T_i \times_{g_i,B} X = T_i \times_{a_i,T} X_T$ and

$$X_{ij} = X_{T_i} \times_{X_T} X_{T_j} = X_i \times_{X_T} X_j$$

with projections pr_i and pr_j to X_i and X_j. Observe that the pullback of (T_i, g_i, E_i) by $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ is given by $(T_i \times_T T_j, g_i \circ \mathrm{pr}_0, L\mathrm{pr}_i^* E_i)$. Hence a descent datum for $\{T_i \to T\}$ in $\mathcal{C}omplexes_{X/B}$ is given by the objects $(T_i, g \circ a_i, E_i)$ and for each pair i, j an isomorphism in $D\mathcal{O}_{X_{ij}}$)

$$\varphi_{ij} : L\mathrm{pr}_i^* E_i \longrightarrow L\mathrm{pr}_j^* E_j$$

satisfying the cocycle condition over the pullback of X to $T_i \times_T T_j \times_T T_k$. Using the vanishing of negative Exts provided by (b) of Lemma 0DLD, we may apply

Simplicial Spaces, Lemma 0DLA to obtain descent[1] for these complexes. In other words, we find there exists an object E in $D_{QCoh}(\mathcal{O}_{X_T})$ restricting to E_i on X_{T_i} compatible with φ_{ij}. Recall that being T-perfect signifies being pseudo-coherent and having locally finite tor dimension over $f^{-1}\mathcal{O}_T$. Thus E is T-perfect by an application of More on Morphisms of Spaces, Lemmas 0DL1 and 0DL2. Finally, we have to check condition (2) from Lemma 0DLD for E. This immediately follows from the description of the open W in Lemma 0DLC and the fact that (2) holds for E_i on X_{T_i}/T_i. □

[1]To check this, first consider the Čech fppf hypercovering $a : T_\bullet \to T$ with $T_n = \coprod_{i_0 \ldots i_n} T_{i_0} \times_T \ldots \times_T T_{i_n}$ and then set $U_\bullet = T_\bullet \times_{a,T} X_T$. Then $U_\bullet \to X_T$ is an fppf hypercovering to which we may apply Simplicial Spaces, Lemma 0DLA.